

The Municipal solid wastes (Management and handling) Rules, 2000

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The Municipal solid wastes (Management and handling) Rules, 2000

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The Municipal solid wastes (Management and handling) Rules, 2000

Rule THE-MUNICIPAL-SOLID-WASTES-MANAGEMENT-AND-HANDLING-RULES-2000 of 2000

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The Municipal solid wastes (Management and handling) Rules, 2000

Vide S.O. 908(E), dated 25.9.2000, published in the Gazette of India, Ext., Point II, Section 3(ii), dated 3.10.2000.

10.

/538

In exercise of the powers conferred by section 3, 6 and 25 of the Environment (Protection) Act, 1986 (

29 of 1986

), the Central Government hereby makes the following rules to regulate the management and handling of the municipal solid wastes, namely :-

1.

Short title and commencement .

-These rules may be called the Municipal Solid Wastes (Management and Handling) Rules, 2000.

(2)

Save as otherwise provided in these rules, they shall come into force on the date of their publication in the Official Gazette.

2.

Application .-

These rules shall apply to every municipal authority responsible for collection, segregation, storage, transportation,, processing and disposal of municipal solid wastes .

3.

Definitions.-

In these rules, unless the context otherwise requires , -

(i)

"anaerobic digestion" means a controlled process involving microbial decomposition of organic matter in the absence of oxygen;

(ii)

"authorization" means the consent given by the Board or Committee to the "operator of a facility" ;

(iii)

"biodegradable substance" means a substance that can be degraded by micro-organisms;

(iv)

"biomethanation" means a process which entails enzymatic decomposition of the organic matter by microbial action to produce methane rich biogas;

(v)

"collection" means lifting and removal of solid wastes from collection points or any other location;

(vi)

"composting" means a controlled process involving microbial decomposition of organic matter;

(vii)

"demolition and construction waste" means wastes from building materials debris and rubble resulting from construction, re-modelling, repair and demolition operation;

(viii)

"disposal" means final disposal of municipal solid wastes in terms of the specified measures to prevent contamination of ground-water, surface water and ambient air quality;

(ix)

"Form" means a Form appended to these rules;

(x)

"generator of wastes" means persons or establishments generating municipal solid wastes;

(xi)

"landfilling" means disposal of residual solid wastes on land in a facility designed with protective measures against pollution of ground water, surface water and air fugitive dust, wind-blown litter, bad odour, fire hazard, bird menace, pests or rodents, greenhouse gas emissions, slope instability and erosion;

(xii)

"leachate" means liquid that seeps through solid wastes or other medium and has extracts of dissolved or suspended material from it;

(xiii)

"lysimeter" is a device used to measure rate of movement of water through or from a soil layer or is used to collect percolated water for quality analysis;

(xiv)

"municipal authority" means Municipal Corporation, Municipality, Nagar Palika, Nagar Nigam, Nagar Panchayat, Municipal Council including notified area committee (NAC) or any other local body constituted under the relevant statutes and, where the management and handling of municipal solid waste is entrusted to such agency;

(xv)

"municipal solid waste" includes commercial and residential wastes generated in a municipal or notified areas in either solid or semi-solid form excluding industrial hazardous wastes but including treated bio-medical wastes;

(xvi)

"operator of a facility" means a person who owns or operates a facility for collection, segregation, storage, transportation, processing and disposal of municipal solid wastes and also includes any other agency appointed as such by the municipal authority for the management and handling of municipal solid wastes in the respective areas;

(xvii)

"pelletisation" means a process whereby pellets are prepared which are small cubes or cylindrical pieces made out of solid wastes and includes fuel pellets which are also referred as refuse derived fuel;

(xviii)

"processing" means the process by which solid wastes are transformed into new or recycled products;

(xix)

"recycling" means the process of transforming segregated solid wastes into raw materials for producing new products, which may or may not be similar to the original products;

(xx)

"Schedule" means a Schedule appended to these rules;

(xxi)

"segregation" means to separate the municipal solid wastes into the groups of organic, inorganic, recyclables and hazardous wastes;

(xxii)

"State Board or the Committee" means the State Pollution Control Board of a State, or as the case may be, the Pollution Control Committee of a Union territory;

(xxiii)

"storage" means the temporary containment of municipal solid wastes in a manner so as to prevent littering, attraction to vectors, stray animals and excessive foul odour;

(xxiiii)

"transportation " means conveyance of municipal solid wastes from place to place hygienically through specially designed transport system so as to prevent foul odour, littering, unsightly conditions and accessibility to vectors;

(xxv)

"vadose water" water which occurs between the ground, surface and the water table that is the unsaturated zone;

(xxvi)

"vermicomposting" is a process of using earthworms for conversion of bio-degradable wastes into compost.

4.

Responsibility of municipal authority .-

Every municipal authority shall, within the territorial area of the municipality, be responsible for the implementation of the provisions of these rules, and for any infrastructure development for collection, storage, segregation, transportation, processing and disposal of municipal solid wastes.

(2)

The municipal authority or an operator of a facility shall make an application in Form-I, for grant of authorization for setting up waste processing and disposal facility including landfills from the State Board or the Committee in order to comply with the implementation programme laid down in Schedule I.

(3)

The municipal authority shall comply with these rules as per the implementation schedule laid down in Schedule I.

(4)

The municipal authority shall furnish its annual report in Form-II,-

(a)

to the Secretary-in-charge of the Department of Urban Development of the concerned State or as the case may be of the Union territory, in case of a metropolitan city; or

(b)

to the District Magistrate or the Deputy Commissioner concerned in case of all other towns and cities, with a copy to the State Board or the Committee on or before the 30th day of June every year.

5.

Responsibility of the State Government and the Union territory Administrations .-

-

(1)

The Secretary-in-charge of the Department of Urban Development of the concerned State or the Union territory, as the case may be, shall have the overall responsibility for the enforcement of the provisions of these rules in the metropolitan cities.

(2)

The District Magistrate or the Deputy Commissioner of the concerned district shall have the overall responsibility for the enforcement of the provisions of these rules within the territorial limits of their jurisdiction.

6.

Responsibility of the Central Pollution Control Board and the State Board or the Committees .-

(1)

The State Board or the Committee shall monitor the compliance of the standards regarding ground water, ambient air, leachate quality and the compost quality including incineration standards as specified under Schedules II, III and IV.

(2)

The State Board or the Committee, after the receipt of application from the municipal authority or the operator of a facility in Form I, for grant of authorization for setting up waste processing and disposal facility including landfills, shall examine the proposal taking into consideration the views of other agencies like the State Urban Development Department, the Town and Country Planning Department, Air Port or Air Base Authority, the Ground Water Board or any such other agency prior to issuing the authorization.

(3)

The State Board or the Committee shall issue the authorization in Form-III to the municipal authority or an operator of a facility within forty-five days stipulating compliance criteria and standards as specified in Schedules II, III and IV including such other conditions, as may be necessary.

(4)

The authorization shall be valid for a given period and after the validity is over, a fresh authorization shall be required.

(5)

The Central Pollution Control Board shall co-ordinate with the State Boards and the Committees with particular reference to implementation and review of standards and guidelines and compilation of monitoring data.

7.

Management of municipal solid wastes .-

(1)

Any municipal solid waste generated in a city or a town, shall be managed and handled in accordance with the compliance criteria and the procedure laid down in Schedule-II.

(2)

The waste processing and disposal facilities to be set up by the municipal authority on their own or through an operator of a facility shall meet the specifications and standards as specified in Schedules III and IV.

8.

Annual Reports .-

(1)

The State Boards and the Committees shall prepare and submit to the Central Pollution Control Board an annual report with regard to the implementation of these rules by the 15th of September every year in Form-IV.

(2)

The Central Pollution Control Board shall prepare the consolidated annual review report on management of municipal solid wastes and forward it to the Central Government alongwith its recommendations before the 15th of December every year.

9.

Accident Reporting .-

When an accident occurs at any municipal solid wastes collection, segregation, storage, processing, treatment and disposal facility or landfill site or during the transportation of such wastes, the municipal authority shall forthwith report the accident in Form-V to the Secretary in-charge of the Urban Development Department in metropolitan cities, and to District Collector or Deputy Commissioner in all other cases.

I

[See rules 4(2) and (3)]

IMPLEMENTATION SCHEDULE

Serial No.

Compliance Criteria

Schedule

1.

Setting up of waste processing and disposal facilities

By 31.12.2003 or earlier

2.

Monitoring the performance of waste processing and disposal facilities

Once in six months

3.

Improvement of existing landfill sites as per provisions of these rules

By 31.12.2001 or earlier

4.

Identification of landfill sites for future use and making site (s) ready for operation

By 31.12.2002 or earlier

II

[See rules 6(1) and (3), 7(1)]

Management of Municipal Solid Wastes

Sl. No

Parameters

Compliance criteria

1.

Collection of municipal solid wastes

1. Littering of municipal solid waste shall be prohibited in cities, towns and in urban areas notified by the State Governments. To prohibit littering and facilitate compliance, the following steps shall be taken by the municipal authority, namely :-

Organising house-to-house collection of municipal solid wastes through any of the methods, like community bin collection (central bin), house-to-house collection, collection on regular pre-informed timings and scheduling by using bell ringing of musical vehicle (without exceeding permissible noise levels);Devising collection of waste from slums and squatter areas or localities including hotels, restaurants, office complexes and commercial areas;Wastes from slaughter houses, meat and fish markets, fruits and vegetable markets, which are biodegradable in nature, shall be managed to make use of such wastes;Bio-medical wastes and industrial wastes shall not be mixed with municipal solid wastes and such wastes shall follow the rules separately specified for the purpose;Collected waste from residential and other areas shall be transferred to community bin by hand-driven containerised carts or other small vehicles;Horticultural and construction or demolition wastes or debris shall be separately collected and disposed off following proper norms. Similarly, wastes generated at dairies shall be regulated in accordance with the State laws;Waste

(garbage, dry leaves) shall not be burnt;Stray animals shall not be allowed to move around waste storage facilities or at any other place in the city or town and shall be managed in accordance with the State laws.2. The municipal authority shall notify waste collection schedule and the likely method to be adopted for public benefit in a city or town.

3. It shall be the responsibility of generator of wastes to avoid littering and ensure delivery of wastes in accordance with the collection and segregation system to be notified by the municipal authority as per para 1(2) of this Schedule

2.

Segregation of municipal solid wastes

In order to encourage the citizens, municipal authority shall organise awareness programmes for segregation of wastes and shall promote recycling or reuse of segregated materials.

The municipal authority shall undertake phased programme to ensure community participation in waste segregation. For this purpose, regular meetings at quarterly intervals shall be arranged by the municipal authorities with representatives of local resident welfare associations and non-governmental organizations.

3.

Storage of municipal solid wastes

Municipal authorities shall establish and maintain storage facilities in such a manner as they do not create unhygienic and insanitary conditions around it. Following criteria shall be taken into account while establishing and maintaining storage facilities, namely :-

Storage facilities shall be created and established by taking into account quantities of waste generation in a given area and the population densities. A storage facility shall be so placed that it is accessible to users; Storage facilities to be set up by municipal authorities or any other agency shall be so designed that wastes stored are not exposed to open atmosphere and shall be aesthetically acceptable and user-friendly; Storage facilities or 'bins' shall have 'easy to operate' design for handling,

transfer and transportation of waste. Bins for storage of bio-degradable wastes shall be painted green, those for storage of recyclable wastes shall be printed white and those for storage of other wastes shall be printed black; Manual handling of waste shall be prohibited. If unavoidable due to constraints, manual handling shall be carried out under proper precaution with due care for safety of workers.

4.

Transportation of municipal solid wastes

Vehicles used for transportation of wastes shall be covered.

Waste should not be visible to public, nor exposed to open environment preventing their scattering. The following criteria shall be met, namely:-

The storage facilities set up by municipal authorities shall be daily attended for clearing of wastes. The bins or containers wherever placed shall be cleaned before they start overflowing; Transportation vehicles shall be so designed that multiple handling of wastes, prior to final disposal, is avoided

5.

Processing of municipal solid wastes

Municipal authorities shall adopt suitable technology or combination of such technologies to make use of wastes so as to minimize burden on landfill. Following criteria shall be adopted, namely:-

(i) The biodegradable wastes shall be processed by composting, vermicomposting, anaerobic digestion or any other appropriate biological processing for stabilization of wastes. It shall be ensured that compost or any other end product shall comply with standards as specified in Schedule-IV;

Mixed waste containing recoverable resources shall follow the route of recycling. Incineration with or without energy recovery including pelletisation can also be used for processing wastes in specific cases.

Municipal authority or the operator of a facility wishing to use other state-of-the-art technologies shall approach the Central Pollution Control Board to get the standards laid down before applying for grant of authorisation.]

6.

Disposal of municipal solid wastes

Land filling shall be restricted to non-biodegradable, inert waste and other waste that are not suitable either for recycling or for biological processing. Land filling shall also be carried out for residues of waste processing facilities as well as pre-processing rejects from waste processing facilities. Land filling of mixed waste shall be avoided unless the same is found unsuitable for waste processing. Under unavoidable circumstances or till installation of alternate facilities, land-filling shall be done following proper norms. Landfill sites shall meet the specifications as given in Schedule -III.

III

[see rules 6(1) and (3), 7(2)]

Specifications for Landfill Sites

Site Selection

1.

In areas falling under the jurisdiction of 'Development Authorities' it shall be the responsibility of such Development Authorities to identify the landfill sites and hand over the sites to the concerned municipal authority for development, operation and maintenance. Elsewhere, this responsibility shall lie with the concerned municipal authority.

2.

Selection of landfill sites shall be based on examination of environmental issues. The Department of Urban Development of the State or the Union territory shall co-ordinate with the concerned organisations for obtaining the

necessary approvals and clearances.

3.

The landfill site shall be planned and designed with proper documentation of a phased construction plan as well as a closure plan.

4.

The landfill sites shall be selected to make use of nearby wastes processing facility. Otherwise, wastes processing facility shall be planned as an integral part of the landfill site.

5.

The existing landfill sites which continue to be used for more than five years, shall be improved in accordance of the specifications given in this Schedule.

6.

Biomedical wastes shall be disposed off in accordance with the Bio-medical Wastes (Management and Handling) Rules, 1998 and hazardous wastes shall be managed in accordance with the Hazardous Wastes (Management and Handling) Rules, 1989, as amended from time to time.

7.

The landfill site shall be large enough to last for 20-25 years.

8.

The landfill site shall be away from habitation clusters, forest areas, water bodies monuments, National Parks, Wetlands and places of important cultural, historical or religious interest.

9.

A buffer zone of no-development shall be maintained around landfill site and shall be incorporated in the Town Planning Department's land-use plans.

10.

Landfill site shall be away from airport including airbase. Necessary approval of airport or airbase authorities prior to the setting up of the landfill site shall be obtained in cases where the site is to be located within 20 km of an airport or airbase..

Facilities at the Site

11.

Landfill site shall be fenced or hedged and provided with proper gate to monitor incoming vehicles or other modes of transportation.

12.

The landfill site shall be well protected to prevent entry of unauthorised persons and stray animals.

13.

Approach and other internal roads for free movement of vehicles and other machinery shall exist at the landfill site.

14.

The landfill site shall have wastes inspection facility to monitor wastes brought in for landfill, office facility for record keeping and shelter for keeping equipment and machinery including pollution monitoring equipments.

15.

Provisions like weigh bridge to measure quantity of waste brought at landfill site, fire protection equipments and other facilities as may be required shall be provided.

16.

Utilities such as drinking water (preferably bathing facilities for workers) and lighting arrangements for easy landfill operations when carried out in night hours shall be provided.

17.

Safety provisions including health inspections of workers at landfill site shall be periodically made.

Specifications for land filling

18.

Wastes subjected to land filling shall be compacted in thin layers using landfill compactors to achieve high density of the wastes. In high rainfall areas where heavy compactors cannot be used alternative measures shall be adopted.

19.

Wastes shall be covered immediately or at the end of each working day with minimum 10 cm of soil, inert debris or

construction material till such time waste processing facilities for composting or recycling or energy recovery are set up as per Schedule I.

20.

Prior to the commencement of monsoon season, an intermediate cover of 40-65 cm thickness of soil shall be placed on the landfill with proper compaction and grading to prevent infiltration during monsoon. Proper drainage berms shall be constructed to divert run-off away from the active cell of the landfill.

21.

After completion of landfill, a final cover shall be designed to minimize infiltration and erosion. The final cover shall meet the following specifications, namely :-

(a)

The final cover shall have a barrier soil layer comprising of 60 cms of clay or amended soil with permeability coefficient less than 1×10^{-7} cm/sec.

(b)

On top of the barrier soil layer there shall be a drainage layer of 15 cm.

(c)

On top of the drainage layer there shall be a vegetative layer of 45 cm to support natural plant growth and to minimize erosion.

Pollution prevention

22.

In order to prevent pollution problems from landfill operations, the following provisions shall be made, namely :-

(a)

Diversion of storm water drains to minimize leachate generation and prevent pollution of surface water and also for avoiding flooding and creation of marshy conditions;

(b)

Construction of a non-permeable lining system at the base and walls of waste disposal area. For landfill receiving residues of waste processing facilities or mixed waste or waste having contamination of hazardous materials (such as aerosols, bleaches, polishes, batteries, waste oils, paint products and pesticides) minimum liner specifications shall be a composite barrier having 1.5 mm high density polyethylene (HDPE) geomembrane, or equivalent, overlying 90 cm of soil (clay or amended soil) having permeability coefficient not greater than 1×10^{-7} cm/sec. The highest level of water table shall be at least two meter below the base of clay or amended soil barrier layer;

(c)

Provisions for management of leachates collection and treatment shall be made. The treated leachates shall meet the standards specified in Schedule- IV;

(d)

Prevention of run-off from landfill area entering any stream, river, lake or pond.

Water Quality Monitoring

23.

Before establishing any landfill site, baseline data of ground water quality in the area shall be collected and kept in record for future reference. The ground water quality within 50 metres of the periphery of landfill site shall be periodically monitored to ensure that the ground water is not contaminated beyond acceptable limit as decided by the Ground Water Board or the State Board or the Committee. Such monitoring shall be carried out to cover different seasons in a year that is, summer, monsoon and post-monsoon period.

24.

Usage of groundwater in and around landfill sites for any purpose (including drinking and irrigation) is to be considered after ensuring its quality. The following specifications for drinking water quality shall apply for monitoring purpose, namely :-

.No.

Parameters

IS 10500: 1991 Desirable limit

(mg/l except for pH)

1.

Arsenic

0.05

2.

Cadmium

0.01

3

Chromium

0.05

4.

Copper

0.05

5.

Cyanide

0.05

6.

Lead

0.05

7.

Mercury

0.001

8.

Nickel

-

9.

Nitrate as NO₃

45.0

10

PH

6.5-8.5

11.

Iron

0.3

12.

Total hardness (as CaCO₃)

300.0

13.

Chlorides

250

14.

Dissolved solids

500

15.

Phenolic compounds (as C₆H₅OH)

0.001

16.

Zinc

5.0

17.

Sulphate (as SO₄)

200

Ambient Air Quality Monitoring

25.

Installation of landfill gas control system including gas collection system shall be made at landfill site to minimize odour generation, prevent off-site migration of gases and to protect vegetation planted on the rehabilitated landfill surface.

26.

The concentration of methane gas generated at landfill site shall not exceed 25 per cent of the lower explosive limit (LEL).

27.

The landfill gas from the collection facility at a landfill site shall be utilized for either direct thermal applications or power generation, as per viability. Otherwise, landfill gas shall be burnt (flared) and shall not be allowed to directly escape to the atmosphere or for illegal tapping. Passive venting shall be allowed if its utilization or flaring is not possible.

28.

Ambient air quality at the landfill site and at the vicinity shall be monitored to meet the following specified standards, namely :-

S.No.

Parameters

Acceptable levels

(i)

Sulphur dioxide

120 ug/m³(24 hours)

(ii)

Suspended Particulate Matter

500 ug/m³(24 hours)

(iii)

Methane

Not to exceed 25 per cent of the lower explosive limit (equivalent) to 650 mg/m³

(iv)

Ammonia daily average

(Sample duration 24 hrs)

0.4 mg/m³(400 mg/m³)

(v)

Carbon monoxide

1 hour average: 2 mg/m³

8 hour average: 1 mg/m³

29.

The ambient air quality monitoring shall be carried out by the concerned authority as per the following schedule, namely:-

(a)

Six times in a year for cities having population of more than fifty lakhs;

(b)

Four times in a year for cities having population between ten and fifty lakhs;

(c)

Two times in a year for town or cities having population between one and ten lakhs.

PLANTATION AT LANDFILL SITE

30.

A vegetative cover shall be provided over the completed site in accordance with the and following specifications, namely :-

(a)

Selection of locally adopted non-edible perennial plants that are resistant to drought and extreme temperatures shall be allowed to grow;

(b)

The plants grown be such that their roots do not penetrate more than 30 cms. This condition shall apply till the landfill is

stabilised;

(c)
Selected plants shall have ability to thrive on low-nutrient soil with minimum nutrient addition;

(d)
Plantation to be made in sufficient density to minimize soil erosion.

CLOSURE OF LANDFILL SITE AND POST-CARE

31.
The post-closure care of landfill site shall be conducted for at least fifteen years and long term monitoring or care plan shall consist of the following, namely :-

(a)
Maintaining the integrity and effectiveness of final cover, making repairs and preventing run-on and run-off from eroding or otherwise damaging the final cover;

(b)
Monitoring leachate collection system in accordance with the requirement;

(c)
Monitoring of ground water in accordance with requirements and maintaining ground water quality;

(d)
Maintaining and operating the landfill gas collection system to meet the standards.

32.
Use of closed landfill sites after fifteen years of post-closure monitoring can be considered for human settlement or otherwise only after ensuring that gaseous and leachate analysis comply with the specified standards.

SPECIAL PROVISIONS FOR HILLY AREAS

33.
Cities and towns located on hills shall have location-specific methods evolved for final disposal of solid wastes by the municipal authority with the approval of the concerned State Board or the Committee. The municipal authority shall set up processing facilities for utilization of biodegradable organic wastes. The inert and non-biodegradable waste shall be used for building roads or filling-up of appropriate areas on hills. Because of constraints in finding adequate land in hilly areas, wastes not suitable for road-laying or filling up shall be disposed of in specially designed landfills.

IV

[See rules 6(1) and (3), 7(2)]

STANDARDS FOR COMPOSTING, TREATED LEACHATES AND INCINERATION

1. The waste processing or disposal facilities shall include composting, incineration, pelletisation, energy recovery or any other facility based on state-of-the-art technology duly approved by the Central Pollution Control Board

2. In case of engagement of private agency by the municipal authority, a specific agreement between the municipal authority and the private agency shall be made particularly, for supply of solid waste and other relevant terms and conditions.

3. In order to prevent pollution problems from compost plant and other processing facilities, the following shall be complied with, namely :-

(i)
The incoming wastes at site shall be maintained prior to further processing. To the extent possible, the waste storage area should be covered. If, such storage is done in an open area, it shall be provided with impermeable base with facility for collection of leachate and surface water run-off into lined drains leading to a leachate treatment and disposal facility;

(ii)
Necessary precautions shall be taken to minimise nuisance of odour, flies, rodents, bird menace and fire hazard;

(iii)
In case of breakdown or maintenance of plant, waste intake shall be stopped and arrangements be worked out for diversion of wastes to the landfill site;

(iv)
Pre-process and post-process rejects shall be removed from the processing facility on regular basis and shall not be allowed to pile at the site. Recyclables shall be routed through appropriate vendors. The non-recyclables shall be sent for well designed landfill site(s).

(v)

In case of compost plant, the windrow area shall be provided with impermeable base. Such a base shall be made of concrete or compacted clay, 50 cm thick, having permeability coefficient less than 10^{-7} cm/sec. The base shall be provided with 1 to 2 per cent slope and circled by lined drains for collection of leachate or surface run-off;

(vi)

Ambient air quality monitoring shall be regularly carried out particularly for checking odour nuisance at down-wind direction on the boundary of processing plant.

(vii)

In order to ensure safe application of compost, the following specifications for compost quality shall be met, namely

Parameters

Concentration not to exceed *

(mg/kg dry basis , except pH value and C/N ratio)

Arsenic

10.00

Cadmium

5.00

Chromium

50.00

Copper

300.00

Lead

100.00

Mercury

0.15

Nickel

50.00

Zinc

1000.00

C/N ratio

20-40

PH

5.5-8.5

* Compost (final product) exceeding the above stated concentration limits shall not be used for food crops. However, it may be utilized for purposes other than growing food crops.

4. The disposal of treated leachates shall follow the following standards, namely:-

Sl.

No

Parameters

Standards

(Mode of Disposal)

Inland surface water

Public sewers

Land disposal

1.

Suspended solids, mg/l, max

100

600

200

2.

Dissolved solids (inorganic) mg/l, max.

2100

2100

2100

3.

PH value

5.5 to 9.0

5.5 to 9.0

5.5 to 9.0

4.

Ammonical nitrogen (as N), mg/l, max.

50

50

-

5.

Total Kjeldahl nitrogen (as N), mg/l, max.

100

-

-

6.

Biochemical oxygen demand (3 days at 270C) max.(mg/l)

30

350

100

7.

Chemical oxygen demand, mg/l, max.

250

-

-

8.

Arsenic (as As), mg/l, max

0.2

0.2

0.2

9.

Mercury (as Hg), mg/l, max

0.01

0.01

-

10.

Lead (as Pb), mg/l, max

0.1

1.0

-

11.

Cadmium (as Cd), mg/l, max

2.0

1.0

-

12.

Total Chromium (as Cr), mg/l, max.

2.0

2.0

-

13.

Copper (as Cu), mg/l, max.

3.0

3.0

-

14.

Zinc (as Zn), mg/l, max.

5.0

15

-

15.

Nickel (as Ni), mg/l, max

3.0

3.0

-

16.

Cyanide (as CN), mg/l, max.

0.2

2.0

0.2

17.

Chloride (as Cl), mg/l, max.

1000

1000

600

18.

Fluoride (as F), mg/l, max

2.0

1.5

-

19.

Phenolic compounds (as C₆H₅OH) mg/l, max.

1.0

5.0

-

Note : While discharging treated leachates into inland surface waters, quantity of leachates being discharged and the quantity of dilution water available in the receiving water body shall be given due consideration.

The incinerators shall meet the following operating and emission standards, namely:-

A. Operating Standards

(1)

The combustion efficiency (CE) shall be at least 99.00%.

(2)

The combustion efficiency is computed as follows

%CO₂

C.E. = ----- x 100

%CO₂ + %CO

Parameters

Concentration mg/Nm³at (12% CO₂correction)

(1)

Particulate matter

150

(2)| Nitrogen Oxides

450|-

(3)| HCl

50|-

(4)| Minimum stack height shall be 30 metres above ground.

|-

(5)| Volatile organic compounds in ash shall not be more than 0.01%

|-

|||

Note : 1. Suitably designed pollution control devices shall be installed or retrofitted with the incinerator to achieve the above emission limits, if necessary.

2. Wastes to be incinerated shall not be chemically treated with any chlorinated disinfectants

3. Chlorinated plastics shall not be incinerated.

4. Toxic metals in incineration ash shall be limited within the regulatory quantities as specified in the Hazardous Wastes (Management and Handling) Rules, 1989 as amended from time to time.

5. Only low sulphur fuel like LDO,LSHSor Diesel shall be used as fuel in the incinerator

Form -I

[See rules 4(2) & 6(2)]

APPLICATION FOR OBTAINING AUTHORIZATION

To,

The Member Secretary

1.

Name of the municipal authority/Name of the agency appointed by the municipal authority

:

2.

Correspondence address

:

Telephone No.

:

Fax No.

:

3.

Nodal Officer & designation(Officer authorised by the municipal authority or agency responsible for operation of processing or disposal facility)

:

4.

Authorization applied for

(Please tick mark)

:

(a) Setting up & operation of waste rocessing facility

(b)Setting up & operation of disposal facility

5.

Detailed proposal of waste processing/disposal facility (to be attached) to include

:

5.1

Processing of Waste

Location of siteName of waste processing technologyDetails of processing technologyQuannitty of waste to be processed per daySite clearance

(from local authority)Details of agreement between municipal authority and operating agencyUtilization programme for waste processed (Product utilization)Methodology for disposal of waste processing rejects (quantity and quality)Measures to be taken for prevention and control of environmental pollutionInvestment on Project and expected returnsMeasures to be taken for safety of workers working in the plant|

5.2

Disposal of Waste

Number of sites indentifiedLayout maps of siteQuantity of waste to be disposed per dayNature and composition of wasteDetails of methodology or criteria followed for site selectionDetails of existing site under operationMethodology and operational details of landfillingMeasures taken to check enviornmental pollution|

Date.....

Signature of Nodal Officer

Form - II

[See rule 4(4)]

FORMAT OF ANNUAL REPORT TO BE SUBMITTED BY THE MUNICIPAL AUTHORITY

(i)

Name of City/Town:.....

(ii)

Population

(iii)

Name of municipal body and address.....

Telephone No. :

Fax :

(iv)

Name of Incharge dealing with.....

municipal solid wastes.....

with designation.....

1.QUANTITY AND COMPOSITION OF SOLID WASTES

(i)

Total quantity of wastes generated per day:

.....

(ii)

Total quantitiy of wasted collected per day:

.....

(iii)

Name of municipal body and address:

.....

(a)Composting:

.....

(b)Vermiculture:

.....

(c)Petllets:

.....

(d)Others, if any, please specify

.....

(iv)

Total quantity of waste disposed by landfilling:

.....

(a)No. of landfull sites used:

.....

(b)Area used:

.....

(c)Whether Weight-bridge facilities available:

Yes

No

(d)Whether area is fenced:

Yes

No

(e)Lighting facility on site:

Yes

No

(f)Whether equipment like Bulldozer, compacters, etc. available.(Please specify)

.....

(g)Total manpower available on site:

.....

(h)Whether covering is done on daily basis:

Yes

No

(i)Whether covering material is used and whether it is adequately available:

.....

(j)Provisions for gas venting provided:

Available

(Yes/No)

Not available

(k)Provision for leachate collection:

Provisions made

Provisions not made

2.

STORAGE FACILITIES

(i)

Area covered for collection of wastes:

.....

(ii)

No. of houses covered:

.....

(iii)

Whether house-to house collection is practised

.....

(if yes, whether done by Municipality or)

.....

Through Private Agency or Non-Governmental

.....

Organisation):

.....

(iv)

Bins:

Specifications

(Shape and size)

Existing Numbers

Proposed for future

(a)RCC Bins

(Capacity):

(b)Trolleys

(Capacity):

(c)Containers

(Capacity):

(d)Dumber Placers:

(e)Others, please specify:

(v)

Whether all bins/collection spots are attended for daily lifting of garbage:

Yes

No

(vi)

Whether lifting of garbage from dustbins is manual or mechanical, i.e., for example by using of front-end loaders

(Please tick mark):

Manual

Loader

Others

(please specify)

3.

Transportation

Existing number

Actually Required/Proposed

(i)

Truck:

(ii)

Truck-Tipper:

(iii)

Tractor-Trailer:

(iv)

Refuse-collector:

(v)

Dumber-placers:

(vi)

Animal Cart:

(vii)

Tricycle:

(viii)

Others

(please specify):

4. Whether any proposal has been made to improve solid wastes management practices.....

5.Are any efforts made to call for private firms etc., to attempt for processing of waste utilizing technologies like:

Waste Utilisation Technology

Proposals

Steps taken (Quantity to be processed)

(i)

Compositing

:

(ii)

Vermiculture

:

(iii)

Pelletisation

:

(iv)

Others if any, please specify

:

6. What provisions are available and how these are implemented to check unhygienic operations of:

(i)

Dairy related activities

:

(ii)

Slaughter houses and unauthorized slaughtering

:

(iii)

Malba

(construction debris) lifting

:

(iv)

Encroachment in Parks, Footpaths etc.

:

7. How many slums are identified and whether these are provided with sanitation facilities:

8.

Are municipal magistrates appointed for taking penal actions:

Yes

No

[If yes, how many cases registered and settled during last three years (give year-wise details)]

9.

Hospital waste management

(i)

How many Hospitals/Clinics under the control of the Corporation

:

(ii)

What methods are followed for disposal of bio-medical wastes?

:

(iii)

Do you have any proposal for setting up of common treatment facility for disposal of bio-medical wastes?

:

(iv)

How many private Nursing Homes, Clinics etc. are operating in the city/town and what steps have been taken to check disposal of their wastes

:

|||||||

.....

Dated.....

Signature of Municipal Commissioner

Form -III

[See-rule 6(2)]

Format for Issue of Authorisation

Date: _____

To,

The Municipal solid wastes (Management and handling) Rules, 2000

Ref: Your application number _____ dt. _____

The _____ State Pollution Control Board/Pollution Control Committee after examining the proposal hereby authorises _____ having their administrative office at _____ to set up and operate waste processing/waste disposal facility at _____ on the terms and conditions (including the standards to comply) attached to this authorization letter.

1. The validity of this authorization is till _____. After the validity, renewal of authorization is to be sought.
2. The _____ State Pollution Control Board/Pollution Control Committees may, at any time, revoke any of the conditions applicable under the authorization and shall communicate the same in writing.
3. Any violation of the provision of the Municipal Solid Wastes (Management and Handling) Rules, 2000 will attract the penal provision of the Environment (Protection) Act, 1986 (29 of 1986).

(Member Secretary)

Date.....

Place.....(Member-Secretary)

State Pollution Control Board/

Pollution Control Committee

Form - IV

[see rule 8(1)]

FORMAT OF ANNUAL REVIEW REPORT TO BE SUBMITTED BY THE STATE POLLUTION CONTROL BOARD/COMMITTEES TO THE CENTRAL POLLUTION CONTROL BOARD

To,
The Chairman,
Central Pollution Control Board,
(Ministry of Environment and Forests)
Government of India,
'Parivesh Bhawan', East Arjun Nagar,
DELHI-110 0032.

1.

Name of the State/Union territory

:

2.

Name & address of the State Pollution Control

:

3.

Board/Pollution Control Committee Number of municipal authorities responsible for management of municipal solid wastes in the State/Union territory under these rules

:

4.

A Summary Statement on progress made by municipal authorities in respect of implementation of Schedule I
[rule 4(3)]

:

Please attach as Annexure-I

5.

A Summary Statement on progress made by municipal authorities in respect of implementation of Schedule II
[rules 6(1) and (3), 7(1)]

:

Please attach as Annexure-II

6.

A Summary Statement on progress made by municipal authorities in respect of implementation of Schedule III
[rules 6(1) and (3), 7(2)]

:

Please attach as Annexure-III

7.

A summary statement on progress made by municipal authorities in respect of implementation of Schedule IV [rules 6(1) and (3), 7(2)]

:

Please attach as Annexure-IV

Date.....

Place.....Chairman or the Member-Secretary

State Pollution Control Board/

Pollution Control Committee

Form - V

[see rule 9]

Accident reporting

1.

Date and time of accident

:

2.

Sequence of events leading to accident

:

3.

The waste involved in accident

:

4.

Assessment of the effects of the accidents on human health and the environment

:

5.

Emergency measures taken

:

6.

Steps taken to alleviate the effects of accidents

:

7.

Steps taken to prevent the recurrence of such an accident

:

Date.....Signature.....

Place.....Designation.....

Related AI tags, queries and research notes

Translation